

D-002 — HCOS™ Human-Centered AI

A Framework for Human-Centered Practice

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Executive Summary

Human-Centered AI is the HCOS™ discipline concerned with designing, implementing, governing, evaluating, and improving artificial intelligence so that it serves people, protects human dignity, and remains subject to meaningful human responsibility.

Artificial intelligence does not operate independently of human systems. It is created, trained, selected, configured, implemented, interpreted, and governed within organizations, communities, professions, and institutions.

The effects of AI therefore depend not only on the technical model, but also on:

- The purpose for which it is used
- The data it receives
- The workflows into which it is introduced
- The people affected by its outputs
- The decisions it is permitted to influence
- The safeguards placed around it
- The quality of human oversight
- The incentives of the organization
- The distribution of power and responsibility
- The system's ability to identify and respond to harm

Within Human-Centered Operating Systems™, Human-Centered AI applies the foundational principles of Human Systems to the design and use of artificial intelligence.

Its purpose is not simply to make AI more user-friendly.

Its purpose is to ensure that human needs, rights, limits, knowledge, judgment, relationships, and long-term well-being remain central throughout the AI lifecycle.

1. Purpose

The Human-Centered AI discipline exists to help people and organizations use artificial intelligence responsibly within real human systems.

Its primary purpose is to ensure that AI is developed and applied in ways that:

- Serve legitimate human purposes
- Protect human dignity
- Support rather than erase human judgment
- Preserve meaningful human oversight
- Strengthen evidence-informed decision-making
- Reduce rather than intensify harmful system burden
- Maintain accountability
- Protect privacy and confidentiality
- Recognize uncertainty and limitations
- Prevent inappropriate automation
- Support accessibility and equitable participation
- Enable learning, correction, and responsible adaptation

Human-Centered AI examines both the capabilities of AI and the conditions under which those capabilities are used.

It asks not only:

What can the AI do?

It also asks:

What should the AI be allowed to do, for whom, under what conditions, with what evidence, and under whose responsibility?

2. Disciplinary Position

An HCOS Foundation states a foundational principle.

An HCOS Discipline defines an ongoing field of knowledge, study, practice, research, and application.

Human-Centered AI is therefore not a single model, safety checklist, governance policy, technical architecture, or implementation method.

It is the broader discipline through which HCOS examines:

- AI design
- AI implementation
- AI-supported decisions
- Human oversight
- Data and evidence integrity
- Workflow integration
- Automation
- Accountability
- AI safety
- Bias
- Transparency
- Validation
- Human–AI collaboration
- AI-related human load
- Social and organizational consequences

The discipline builds upon **D-001 – HCOS™ Human Systems**.

Human Systems provides the foundational understanding of people, structures, workflows, power, relationships, and system conditions. Human-Centered AI applies that understanding to technologies that generate, classify, predict, recommend, summarize, automate, or support decisions.

3. Discipline Definition

HCOS™ Human-Centered AI is the discipline concerned with designing, implementing, governing, evaluating, and improving artificial intelligence so that it supports legitimate human purposes, protects dignity, preserves accountability, and remains appropriately integrated within the human systems it affects.

4. Why This Discipline Exists

Artificial intelligence is increasingly involved in activities that affect human lives.

AI may help:

- Organize information
- Generate text or images

- Identify patterns
- Summarize evidence
- Predict outcomes
- Recommend actions
- Prioritize work
- Support diagnosis
- Screen applicants
- Personalize learning
- Detect risk
- Allocate resources
- Automate administrative tasks
- Influence professional or organizational decisions

These capabilities may create meaningful benefits.

They may also create harm when AI is introduced without sufficient attention to the surrounding human system.

Common failures may include:

- Automating a flawed process
- Treating incomplete data as complete
- Presenting probability as certainty
- Hiding responsibility behind an algorithm
- Increasing workload through poorly designed implementation
- Replacing human contact where it remains necessary
- Using AI in decisions that require contextual judgment
- Creating systems that cannot be meaningfully challenged
- Allowing biased or unrepresentative data to shape outcomes
- Exposing confidential information
- Using productivity gains to intensify expectations rather than reduce burden
- Measuring what is easy rather than what is important
- Treating AI output as evidence without verification
- Introducing technology without listening to the people affected

Human-Centered AI exists because technical capability alone does not determine whether an AI application is responsible, safe, useful, or humane.

AI must be understood as part of a wider sociotechnical system.

5. The Human Problem

People are increasingly affected by AI-supported systems they may not understand, control, or have the ability to challenge.

They may experience:

- Decisions made without adequate context
- Loss of agency
- Reduced access to human review
- Increased surveillance
- Inaccurate classifications
- Biased outcomes
- Greater administrative burden
- Confusing or opaque processes
- Decreased professional autonomy
- Pressure to trust unreliable outputs
- Reduced privacy
- Difficulty identifying who is accountable
- Exclusion from decisions about technologies that affect them

Workers may also be expected to compensate for poorly implemented AI.

They may need to:

- Correct inaccurate outputs
- Verify generated information
- Re-enter information across systems
- Explain AI decisions to others
- Manage new alerts
- Repair disrupted workflows
- Monitor automated processes
- Carry responsibility without receiving authority
- Absorb increased productivity expectations

In these cases, AI may appear efficient at the organizational level while transferring additional work, risk, or uncertainty to people.

Human-Centered AI seeks to make these transfers visible.

In Simple Terms

AI is like a powerful tool.

A tool can help people build something safely, but only when:

- It is the right tool for the job
- People understand how to use it
- Someone checks whether it is working correctly
- The tool has safety protections
- People can stop using it when something goes wrong

A powerful tool should not be placed in charge of people simply because it can work quickly.

Human-Centered AI makes sure that people remain responsible for how the tool is designed, used, checked, and corrected.

6. Scope

Human-Centered AI includes the study and application of human-centered principles across:

- Generative AI
- Predictive AI
- Machine learning
- Natural language processing
- Decision-support systems
- Recommendation systems
- Automation systems
- Conversational AI
- AI agents
- Computer vision
- Risk-scoring systems
- Classification systems
- Robotics
- AI-supported research
- AI-supported education
- AI-supported healthcare
- AI-supported employment systems
- AI-supported public services
- AI-supported organizational operations

The discipline examines the complete AI lifecycle:

1. Problem definition
2. Purpose determination
3. Evidence gathering
4. Data selection
5. Model design or selection
6. Development
7. Testing
8. Validation
9. Procurement
10. Workflow integration
11. Implementation
12. Human training
13. Use
14. Oversight
15. Monitoring
16. Incident response

- 17. Evaluation
- 18. Revision
- 19. Retirement

Human-Centered AI also examines the organizational structures, professional roles, policies, incentives, and cultural conditions that influence how AI is used.

7. Guiding Foundations

Human-Centered AI is guided by the HCOS Foundations and the principles established within D-001 — Human Systems.

Core propositions include:

- Every organization is a human system.
- Human systems exist to serve people.
- Human dignity is inherent.
- Systems shape human behavior and outcomes.
- Humans and AI must steward one another responsibly.
- Healthy systems create conditions for human flourishing.
- Healthy systems learn, adapt, and steward the future.
- Knowledge begins with evidence.
- Human suffering requires meaningful system stopping points.
- People should not be required to carry unlimited system load.
- Technology should remain accountable to legitimate human purposes.
- Responsibility should remain with people and institutions capable of exercising power.

These foundations shape how AI should be evaluated.

An AI system should not be considered human-centered merely because it is easy to use or preferred by users.

Human-centered evaluation must also consider:

- Human dignity
 - Safety
 - Accountability
 - Evidence quality
 - Fairness
 - Accessibility
 - Privacy
 - Human load
 - Power
 - Oversight
 - Long-term consequences
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8. Core Principles

8.1 AI Must Serve a Legitimate Human Purpose

AI should be introduced only when the purpose is clearly defined and the proposed use meaningfully serves people or the systems upon which they depend.

The existence of a technical capability does not create an obligation to use it.

Questions include:

- What human or system problem is being addressed?
 - Who defined the problem?
 - Who is expected to benefit?
 - Who may be harmed?
 - Is AI necessary?
 - Could a simpler solution work better?
 - Does the application align with the organization's stated values?
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8.2 Human Dignity Must Not Be Automated Away

People should not be reduced to predictions, scores, categories, productivity measurements, risks, or data points.

AI-supported systems should preserve:

- Respect
- Agency
- Context
- Meaningful participation
- Access to human review
- The ability to question consequential decisions
- Recognition of the limits of data

A system may be technically accurate in aggregate while still treating individual people unfairly or without dignity.

8.3 Responsibility Must Remain Human

AI may generate or recommend an output.

It cannot assume moral, professional, legal, organizational, or relational responsibility for the consequences of that output.

Responsibility remains with the people and institutions that:

- Design the system
- Select the technology
- Approve its use
- Configure its authority
- Interpret its outputs
- Act upon its recommendations
- Monitor its performance
- Respond to harm

Statements such as “the algorithm decided” should not be used to conceal human responsibility.

8.4 Oversight Must Be Meaningful

Human oversight is not meaningful merely because a person is present somewhere in the process.

Meaningful oversight requires that the reviewer has:

- Sufficient knowledge
- Access to relevant evidence
- Time to review
- Authority to disagree
- Ability to stop or override the process
- Clear responsibility
- Protection from retaliation
- A practical escalation pathway

A human who must approve hundreds of AI outputs without enough time or authority is not providing meaningful oversight.

8.5 AI Should Support, Not Degrade, Human Judgment

AI may help people organize evidence, identify patterns, compare options, or recognize missing information.

It should not create conditions in which humans become passive, overdependent, or unable to exercise professional judgment.

Human-Centered AI should protect against:

- Automation bias
- Deskilling
- Overreliance

- False confidence
 - Loss of contextual understanding
 - Erosion of professional responsibility
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8.6 Evidence Must Be Verified

AI-generated content is not automatically reliable evidence.

Outputs may contain:

- Errors
- Fabricated citations
- Unsupported conclusions
- Missing context
- Outdated information
- Biased assumptions
- Inappropriate generalizations

AI-supported knowledge work must include verification appropriate to the level of risk.

High-consequence decisions require stronger evidence, documentation, and review than low-risk exploratory tasks.

8.7 Uncertainty Must Be Visible

AI systems should not present uncertain, probabilistic, incomplete, or inferred information as established fact.

Users should be able to understand:

- What the system knows
- What it does not know
- What it inferred
- What evidence was used
- Where uncertainty exists
- What limitations apply
- When human review is required

Confidence should not be confused with correctness.

8.8 Human Load Must Be Evaluated

AI implementation should examine how work changes for people.

This includes whether AI:

- Reduces or increases cognitive load
- Adds verification work
- Creates new documentation requirements
- Produces excessive alerts
- Fragments workflows
- Intensifies productivity expectations
- Shifts work to less powerful groups
- Removes recovery time
- Creates moral or emotional burden

Efficiency should not be declared solely because the AI performs a task quickly.

The complete human and system workload must be examined.

8.9 Privacy and Confidentiality Must Be Protected

AI systems should use only the information necessary for the intended purpose.

Organizations should establish appropriate controls for:

- Data collection
- Data access
- Data sharing
- Storage
- Retention
- De-identification
- Confidentiality
- Security
- Consent
- Secondary use
- Vendor access

Sensitive information should not be entered into AI systems without appropriate authorization and safeguards.

8.10 Fairness Requires Ongoing Examination

AI fairness cannot be established through a single test.

It requires continued examination of:

- Data representation
- Error distribution
- Access
- usability
- Cultural assumptions

- Language
- Disability
- socioeconomic conditions
- Historical inequities
- Differential impact
- Appeal processes
- Human review

A system that performs well overall may still create unacceptable harm for particular groups.

8.11 AI Systems Must Be Able to Stop

Human-Centered AI requires meaningful stopping points.

An AI system should be paused, limited, overridden, or retired when:

- Harm is detected
- Evidence becomes unreliable
- Performance changes
- Data conditions shift
- Oversight fails
- Users cannot safely interpret outputs
- The system is used beyond its intended purpose
- Regulatory or ethical concerns emerge
- The burden outweighs the benefit

Continued operation should not be treated as the default when safety is uncertain.

8.12 Humans and AI Must Steward One Another

Human stewardship of AI requires people to:

- Define responsible purposes
- Protect data
- Establish boundaries
- Verify outputs
- Monitor performance
- Correct errors
- Prevent misuse
- Maintain accountability

AI stewardship of human work may include helping people:

- Organize complex information
- Identify missing evidence
- Recognize patterns

- Reduce avoidable administrative burden
- Support learning
- Improve accessibility
- Explore alternatives
- Preserve important knowledge

This relationship must remain reciprocal but not equal in responsibility.

Humans and human institutions retain final accountability.

9. Primary Areas of Study

9.1 AI Purpose and Appropriateness

This area examines whether AI is an appropriate response to the identified problem.

Questions include:

- Is the problem clearly defined?
 - Is AI necessary?
 - What alternatives exist?
 - What benefits are expected?
 - What risks are introduced?
 - What should remain human?
 - What should never be automated?
 - How will success be defined?
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9.2 Human–AI Interaction

Human–AI Interaction examines how people understand, use, interpret, challenge, and collaborate with AI systems.

Topics include:

- Usability
- Accessibility
- Trust
- Reliance
- Explanations
- Feedback
- Error recovery
- Escalation
- User control
- Cognitive load
- Professional judgment

- Human agency
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9.3 AI Decision Support

AI Decision Support examines systems that inform or influence decisions.

This includes:

- Recommendations
- Risk scores
- Predictions
- Prioritization
- Classification
- Triage
- Resource allocation
- Diagnostic support
- Hiring support
- Educational support
- Policy analysis

The discipline distinguishes among:

- AI-generated information
- AI-supported recommendations
- AI-influenced decisions
- Human decisions
- Automated decisions

The level of oversight should increase as the consequence of the decision increases.

9.4 Responsible Automation

Responsible Automation examines which tasks may be automated and under what conditions.

Key questions include:

- Is the task stable and well-defined?
 - Are exceptions common?
 - Does the task require empathy or contextual judgment?
 - What happens when automation fails?
 - Can people intervene?
 - Does automation shift burden elsewhere?
 - Will people retain necessary skills?
 - Is there a safe manual alternative?
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9.5 AI Governance

AI Governance examines the structures through which AI decisions are authorized, monitored, reviewed, and corrected.

It may include:

- Governance bodies
 - Decision rights
 - Use-case approval
 - Risk classification
 - Procurement review
 - Model inventories
 - Documentation
 - Data governance
 - Validation requirements
 - Incident reporting
 - Change management
 - Appeals
 - Retirement criteria
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9.6 Human Oversight

Human Oversight examines the roles, knowledge, authority, workload, and protections required for responsible review.

It considers:

- Who reviews the output?
 - What must be verified?
 - What qualifications are needed?
 - What level of review is proportionate?
 - Who may override the system?
 - How are disagreements handled?
 - Who is accountable for the final decision?
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9.7 AI Safety and Risk

AI Safety and Risk examines potential harm across technical, human, organizational, and social dimensions.

Risk areas may include:

- Inaccuracy
- Hallucination
- Bias

- Privacy loss
 - Security vulnerabilities
 - Misuse
 - Overreliance
 - Workflow disruption
 - Human load
 - Loss of access
 - Lack of contestability
 - Reputational harm
 - Legal or regulatory risk
 - Professional boundary violations
 - Unintended system behavior
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9.8 Transparency and Explainability

Transparency concerns whether people can understand how AI is being used within a system.

Explainability concerns whether relevant aspects of an AI-supported output can be interpreted sufficiently for the decision at hand.

Human-Centered AI asks:

- Are people told when AI is being used?
 - Can the purpose be explained?
 - Can the output be challenged?
 - Is the evidence traceable?
 - Are limitations visible?
 - Is the explanation understandable to the affected person?
 - Is the explanation sufficient for the level of consequence?
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9.9 Evidence and Data Integrity

This area examines the quality, relevance, completeness, provenance, and protection of the data and evidence used by AI.

Topics include:

- Data quality
- Data representation
- Missing data
- Label quality
- Data provenance
- Evidence currency
- Citation verification
- Context preservation
- Data drift

- Model drift
 - Secondary use
 - Synthetic data
 - Documentation
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9.10 AI-Related Human Load

This area examines the cognitive, emotional, administrative, relational, and professional demands created by AI-supported systems.

Examples include:

- Verification burden
 - Alert burden
 - Exception handling
 - Documentation burden
 - Workflow switching
 - Monitoring requirements
 - Responsibility without authority
 - Pressure to accept AI output
 - Emotional burden from harmful decisions
 - Loss of professional meaning
 - Constant adaptation to new tools
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9.11 AI Education and Literacy

AI Education and Literacy examines what people need to understand in order to use, evaluate, govern, or be affected by AI responsibly.

Competency areas may include:

- Basic AI concepts
 - Capabilities and limitations
 - Evidence verification
 - Bias recognition
 - Privacy awareness
 - Prompt design
 - Human oversight
 - Risk recognition
 - Appropriate use
 - Escalation
 - Ethical reasoning
 - Documentation
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9.12 AI Evaluation and Learning

This area examines how AI-supported systems are monitored after implementation.

Evaluation may consider:

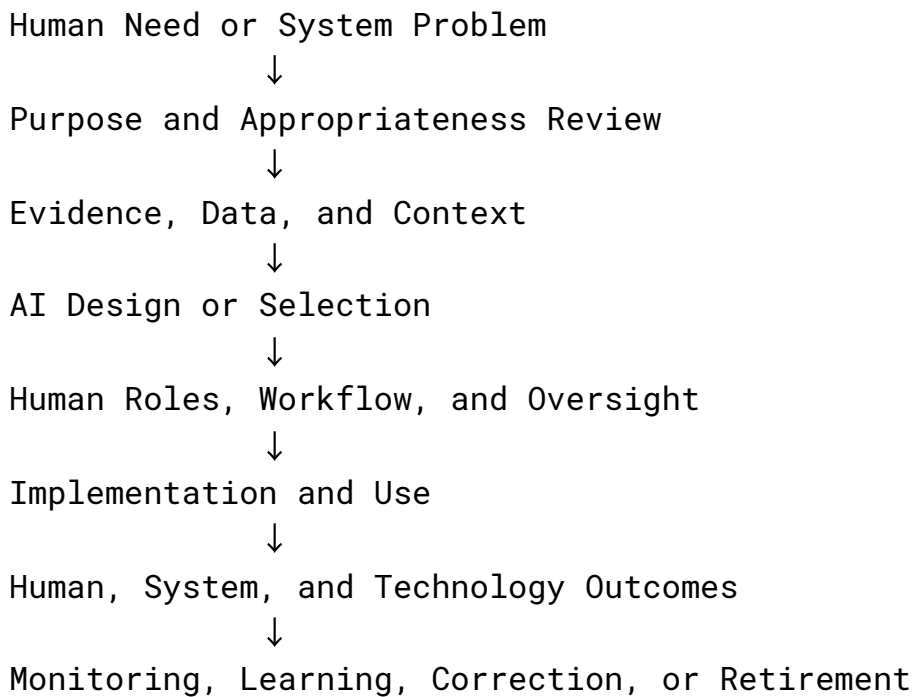
- Accuracy
- Reliability
- Human outcomes
- Workflow outcomes
- Equity
- Safety
- Human load
- Accessibility
- Trust
- Adoption
- Unintended consequences
- Override patterns
- Incident frequency
- Evidence quality
- Long-term sustainability

Implementation is not the end of AI governance.

It is the beginning of ongoing evaluation.

10. A Human-Centered AI Framework

Human-Centered AI may be examined through the following relationship:



Each layer must remain connected.

A technically capable AI system may still be inappropriate when:

- The purpose is poorly defined
- The evidence is weak
- The workflow is unsafe
- Oversight is symbolic
- Human burden increases
- Responsibility is unclear
- People cannot challenge the output
- The system cannot be stopped

11. Human-Centered AI Practice Journey

Stage 1: Identify the Human or System Need

Purpose:

Define the actual problem before selecting a technology.

Key questions:

- What is happening?
- Who is affected?

- What evidence supports the problem definition?
- Is the problem technical, structural, relational, operational, or mixed?
- Is AI necessary?

Common failure:

Beginning with an AI product and searching for a problem it can solve.

Stage 2: Assess Appropriateness

Purpose:

Determine whether AI is suitable for the task.

Key questions:

- What level of consequence is involved?
- Is the task predictable?
- Are exceptions common?
- Is human judgment essential?
- What should remain human?
- What simpler alternatives exist?

Common failure:

Assuming automation is automatically better.

Stage 3: Gather Evidence and Context

Purpose:

Understand the evidence, data, affected populations, workflows, and system conditions.

Key questions:

- What evidence is available?
- Whose knowledge is missing?
- Are the data representative?
- What existing burdens are present?
- What legal, ethical, professional, or regulatory requirements apply?

Common failure:

Treating available data as complete reality.

Stage 4: Design Governance and Safeguards

Purpose:

Define authority, responsibility, oversight, escalation, and stopping conditions.

Key questions:

- Who owns the use case?
- Who reviews outputs?
- Who may override the system?
- How will incidents be reported?
- What information must be documented?
- When must use stop?

Common failure:

Assigning oversight after implementation rather than designing it from the beginning.

Stage 5: Design the Human–AI Workflow

Purpose:

Integrate AI into actual work without fragmenting responsibility or increasing hidden burden.

Key questions:

- Where does AI enter the workflow?
- What must humans verify?
- How much time will verification require?
- What exceptions will occur?
- What manual alternative exists?
- How will users provide feedback?

Common failure:

Evaluating only the AI task rather than the complete human workflow.

Stage 6: Test and Validate

Purpose:

Examine technical performance, usability, human outcomes, and risk before broader implementation.

Key questions:

- Does it work for the intended population?
- How are errors distributed?

- Can users understand and challenge outputs?
- Does human load increase?
- Are safeguards functioning?
- What limitations must be disclosed?

Common failure:

Equating technical accuracy with safe implementation.

Stage 7: Implement Responsibly

Purpose:

Introduce the AI system with training, communication, support, and clear boundaries.

Key questions:

- Do users understand the intended use?
- Are limitations clear?
- Are people prepared to identify failure?
- Is support available?
- Can affected people request human review?
- Is implementation reversible?

Common failure:

Treating training as a one-time technical demonstration.

Stage 8: Monitor Outcomes

Purpose:

Evaluate what happens after implementation.

Key questions:

- Is the system producing the intended benefit?
- What harms or burdens are emerging?
- Are people overriding outputs?
- Has data or workflow changed?
- Are some groups affected differently?
- Are users adapting in unsafe ways?

Common failure:

Monitoring system performance while ignoring human experience.

Stage 9: Learn, Correct, or Retire

Purpose:

Respond to evidence and changing conditions.

Key questions:

- What needs correction?
- Should the scope be limited?
- Should the system be paused?
- Is retraining necessary?
- Has the technology outlived its purpose?
- Is continued use still justified?

Common failure:

Keeping an AI system in use because the organization has already invested in it.

12. Responsibility Must Follow Power

The people most affected by an AI system should have meaningful opportunities to participate in its design and evaluation.

They should not, however, be made solely responsible for correcting harms created by organizations with greater authority and resources.

Responsibility should align with the power to act.

This includes:

- Leaders who approve AI strategy
- Organizations that purchase or deploy AI
- Developers who design systems
- Vendors that make capability and performance claims
- Data stewards who control information
- Professionals who act upon AI-supported recommendations
- Governance bodies that establish boundaries
- Regulators and oversight institutions
- Human reviewers responsible for consequential decisions

A person affected by an AI decision should not be required to understand the entire technical system in order to receive fair treatment or request review.

13. Methods Commonly Used

Human-Centered AI draws from multiple established and emerging fields, including:

- Human-centered design
- Human factors engineering
- Systems thinking
- Sociotechnical systems theory
- AI ethics
- Data science
- Machine learning evaluation
- Safety science
- Implementation science
- Decision science
- Organizational psychology
- Privacy engineering
- Cybersecurity
- Quality improvement
- Participatory design
- Accessibility design
- Risk management
- Evidence-based practice
- Technology governance

Common methods may include:

- AI use-case assessment
- Human-centered risk assessment
- Workflow mapping
- Human oversight design
- Data and evidence review
- Algorithmic impact assessment
- Bias and equity review
- Privacy review
- Human load assessment
- Failure-mode analysis
- Scenario testing
- Red-team review
- Usability testing
- Accessibility testing
- Pilot implementation
- Incident analysis
- Outcome monitoring
- Model and system documentation
- Post-implementation review

HCOS does not replace technical, legal, regulatory, clinical, privacy, security, or ethical expertise.

It integrates those areas within a coherent human-centered system perspective.

14. Human-Centered AI Within HCOS™

HCOS Discipline	Relationship to Human-Centered AI
D-001 Human Systems	Provides the foundational understanding of people, system conditions, workflow, power, human load, learning, and flourishing.
D-003 Healthcare Systems	Applies Human-Centered AI principles to healthcare access, clinical care, medication systems, patient safety, workforce conditions, and health technology.
D-004 Knowledge Stewardship	Protects evidence integrity, data provenance, documentation quality, uncertainty, and responsible knowledge use within AI-supported systems.
D-005 Professional Identity	Examines how AI affects professional roles, judgment, career development, recognition, autonomy, and identity.
D-006 Learning and Education	Develops AI literacy, responsible use competencies, educational applications, and learning systems.
D-007 Leadership and Governance	Establishes authority, accountability, oversight, policy, risk ownership, decision rights, and institutional stewardship.
D-008 Research and Innovation	Develops and evaluates AI models, instruments, implementation methods, interventions, and evidence.

Human-Centered AI should not operate as an isolated technical discipline.

It requires continuous collaboration with every other HCOS discipline.

15. Typical Questions

Human-Centered AI practitioners may ask:

- What human or system problem is this AI intended to address?
 - Is AI necessary for this purpose?
 - Who benefits?
 - Who carries the risk?
 - Who has the authority to decide?
 - What should remain human?
 - What evidence supports the AI's use?
 - What data are being used?
 - Whose experiences are missing from the data?
 - How will uncertainty be communicated?
 - Can people challenge or appeal the output?
 - Who verifies the result?
 - Does the reviewer have enough time and authority?
 - How will the AI affect human workload?
 - Could it increase pressure or surveillance?
 - What happens when it is wrong?
 - How will harm be detected?
 - Can the system be stopped?
 - Who remains accountable?
 - How will the organization learn and adapt?
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16. Intended Outcomes

Human-Centered AI work is intended to contribute to:

- More responsible AI design
- Clearer AI purpose
- Stronger human oversight
- Better evidence verification
- Improved accountability
- Safer workflow integration
- Reduced inappropriate automation
- Better protection of privacy
- Greater transparency
- More accessible systems
- Earlier recognition of bias and harm
- Reduced hidden human burden
- Improved human–AI collaboration
- Stronger AI literacy
- More trustworthy decision support
- Responsible innovation
- Better alignment between technology and human needs

These are intended outcomes rather than guaranteed effects.

Specific outcomes must be evaluated within the setting, population, workflow, and decision context in which an AI system is used.

17. Boundaries and Limitations

Human-Centered AI is interdisciplinary, but it does not replace specialized expertise.

AI-related work may require review by professionals in:

- Computer science
- Data science
- Machine learning
- Cybersecurity
- Privacy
- Law
- Ethics
- Healthcare
- Pharmacy
- Nursing
- Medicine
- Education
- Human resources
- Accessibility
- Public policy
- Safety science
- Research methodology
- Regulatory affairs

The discipline should not be used to:

- Declare an AI system safe without appropriate testing
- Establish legal or regulatory compliance without qualified review
- Replace clinical judgment
- Make autonomous high-consequence decisions without sufficient safeguards
- Treat AI output as verified evidence
- Infer sensitive personal information without justification
- Use human-centered language to legitimize harmful surveillance
- Claim that bias has been permanently eliminated
- Assume that human oversight automatically makes a system safe
- Automate decisions simply because automation is technically possible
- Conceal responsibility behind proprietary technology
- Introduce AI without evaluating human workload
- Present conceptual HCOS instruments as validated tools without evidence

Human-Centered AI cannot eliminate all risk.

Its purpose is to make risks, responsibilities, assumptions, and human consequences more visible and manageable.

18. Risks and Potential Misuse

Human-Centered Language Without Structural Change

Organizations may describe AI as human-centered while leaving power, workload, surveillance, or accountability unchanged.

Mitigation:

Require evidence of participation, oversight, impact evaluation, and corrective authority.

Symbolic Human Oversight

A human may be assigned to review AI outputs without adequate time, knowledge, or authority.

Mitigation:

Define minimum oversight conditions and evaluate actual review capacity.

Automation Bias

People may accept AI outputs because the system appears authoritative or technically advanced.

Mitigation:

Provide training, uncertainty disclosures, independent verification, and protected override authority.

AI Washing

Organizations may overstate the sophistication, effectiveness, fairness, or safety of an AI system.

Mitigation:

Require transparent documentation, evidence review, independent testing, and careful limitation of claims.

Workload Transfer

AI may reduce visible work in one area while increasing verification, correction, monitoring, or exception handling elsewhere.

Mitigation:

Evaluate the complete workflow and human load before and after implementation.

Surveillance Expansion

AI may be used to monitor workers, patients, students, or communities beyond what is necessary or proportionate.

Mitigation:

Establish purpose limitations, data minimization, governance review, transparency, and meaningful challenge processes.

Loss of Professional Judgment

Overreliance on AI may weaken human expertise and confidence.

Mitigation:

Preserve human decision rights, reflective review, competency development, and opportunities to practice independent judgment.

Responsibility Diffusion

Organizations may use AI to obscure who made or authorized a decision.

Mitigation:

Document decision ownership, approval authority, human review, and final accountability.

Premature Claims of Validation

A pilot, demonstration, or limited evaluation may be presented as broad evidence of safety or effectiveness.

Mitigation:

Clearly distinguish conceptual, experimental, piloted, validated, and operationally proven systems.

19. Related HCOS Documents

Foundations

- HCOS-F-001 through HCOS-F-009
- Additional Foundations as formally established

Disciplines

- D-001 — HCOS™ Human Systems
- D-003 — HCOS™ Healthcare Systems
- D-004 — HCOS™ Knowledge Stewardship
- D-005 — HCOS™ Professional Identity
- D-006 — HCOS™ Learning and Education
- D-007 — HCOS™ Leadership and Governance
- D-008 — HCOS™ Research and Innovation

Architectures

- Human-Centered AI Architecture
- Human Systems Architecture
- Healthcare Systems Architecture
- Professional Identity AI Infrastructure

Standards

- HCOS-AI-001 — Human-Centered AI
- AI Governance
- AI Readiness
- AI Implementation
- AI-Assisted Communication
- AI Decision Support
- AI Documentation
- AI Workflow Design
- Human Oversight
- AI Validation
- AI Safety and Risk
- AI Transparency
- AI Education and Training
- Healthcare AI
- Educational AI

- Leadership and AI
- Responsible Automation

Codes and titles should be updated as the official HCOS Standards nomenclature is finalized.

Knowledge Governance

- HCOS™ Annotated Evidence Repository
- HCOS™ Annotated Evidence Records
- Knowledge Governance documents KG-001 through KG-008
- Evidence validation and provenance procedures
- AI-assisted evidence review guidance

Methods and Models

- Human-Centered AI Review
- AI Use-Case Assessment
- Human Oversight Model
- Human Load Assessment
- Pressure and Recovery Model
- 8 + 3 + 1 Human-Centered System Model
- Human-Centered System Mapping
- AI Implementation Journey
- AI Decision Intelligence Model

20. Priority Evidence Areas

The Human-Centered AI evidence base may include research and guidance concerning:

- Human-AI collaboration
- Automation bias
- Trust in AI
- AI explainability
- Algorithmic fairness
- Human factors
- Cognitive load
- Decision support
- Sociotechnical systems
- Implementation science
- AI governance
- Privacy
- Cybersecurity
- Professional judgment
- Workflow integration
- Safety science

- Data quality
- Organizational learning
- AI literacy
- Accessibility
- Human oversight
- Burnout and technology burden

Evidence should be distinguished among:

- Evidence that a problem exists
 - Evidence explaining mechanisms
 - Evidence supporting an intervention
 - Evidence of technical performance
 - Evidence of implementation feasibility
 - Evidence of human outcomes
 - Evidence of system outcomes
 - Evidence of unintended consequences
-

21. Future Development

The Human-Centered AI discipline will continue to develop through:

- Interdisciplinary evidence review
- Technical review
- Legal and regulatory review
- Ethical review
- Privacy and security review
- Professional review
- Practitioner consultation
- Lived-experience participation
- Public and community feedback
- Pilot testing
- Case study development
- Instrument development
- Reliability and validity testing where appropriate
- Implementation research
- Cross-cultural review
- Accessibility review
- AI safety testing
- Version-controlled revision

Priority future work includes:

1. Establishing the formal HCOS Human-Centered AI Architecture.
2. Developing an AI use-case classification model.
3. Defining minimum requirements for meaningful human oversight.

4. Creating a Human-Centered AI Implementation Standard.
 5. Developing an AI-related Human Load Assessment.
 6. Establishing a standardized AI risk and impact review process.
 7. Building a Human-Centered AI evidence map.
 8. Developing healthcare, education, employment, government, and organizational case studies.
 9. Creating public and professional AI literacy pathways.
 10. Defining AI stopping, correction, and retirement criteria.
 11. Developing governance templates and decision logs.
 12. Establishing a formal Human-Centered AI research agenda.
-

22. Conclusion

Human-Centered AI is the HCOS discipline through which artificial intelligence is understood as part of a larger human system.

It recognizes that AI is never only a technical product.

It is also a set of human choices involving purpose, power, evidence, responsibility, workflow, risk, and consequence.

The goal of Human-Centered AI is not to prevent innovation.

It is to ensure that innovation remains accountable to the people and communities it affects.

AI should help people think, learn, decide, create, and care more effectively without requiring them to surrender dignity, agency, judgment, privacy, or meaningful human connection.

Human-Centered AI begins with a clear responsibility:

Artificial intelligence should serve humanity without becoming a mechanism through which human responsibility disappears.

Discipline at a Glance

Discipline:

HCOS™ Human-Centered AI

Code:

D-002

Primary concern:

Designing, implementing, governing, and evaluating artificial intelligence so that it serves people and remains subject to meaningful human responsibility.

Human problem addressed:

People are increasingly affected by AI-supported systems that may be inaccurate, opaque, burdensome, difficult to challenge, or disconnected from human needs and context.

Core system question:

How can artificial intelligence support human capability and system performance while protecting dignity, agency, safety, evidence integrity, and accountability?

Primary users:

- AI developers
- AI product teams
- Organizational leaders
- Governance bodies
- Healthcare professionals
- Educators
- Researchers
- Policy professionals
- Privacy and security professionals
- Human factors specialists
- AI implementation teams
- Human reviewers
- Communities affected by AI

Primary outputs:

- AI use-case assessments
- Governance frameworks
- Human oversight plans
- Risk assessments
- Workflow designs
- Evidence reviews
- Implementation plans
- Monitoring frameworks
- Educational resources
- AI standards
- Decision logs
- Human-centered AI tools and instruments

Key safeguards:

- Human dignity
- Legitimate purpose

- Meaningful human oversight
- Evidence verification
- Transparency
- Privacy
- Security
- Accessibility
- Human accountability
- Sustainable human load
- Challenge and appeal processes
- Meaningful stopping points

Related HCOS disciplines:

- D-001 Human Systems
- D-003 Healthcare Systems
- D-004 Knowledge Stewardship
- D-005 Professional Identity
- D-006 Learning and Education
- D-007 Leadership and Governance
- D-008 Research and Innovation

Development status:

Version 1.0 – Draft

Document Governance

Document Status

This document is a developmental framework published as **Version 1.0 – Draft**.

It is intended to support discussion, education, research, structured reflection, interdisciplinary collaboration, governance development, and responsible AI practice.

Evidence Status

This document includes:

- Ethical principles
- Conceptual propositions
- Practice-informed observations
- Emerging areas of professional consensus
- Topics supported to varying degrees by existing evidence
- Research questions requiring further investigation

Conceptual statements should not be presented as established scientific findings unless supported by appropriate evidence.

Human Review

Human review remains necessary before this discipline or its related models, standards, methods, instruments, or AI applications are used to support consequential:

- Clinical decisions
- Employment decisions
- Legal decisions
- Educational decisions
- Financial decisions
- Safety decisions
- Public policy decisions
- Organizational decisions
- Resource-allocation decisions

AI Disclosure

This document was developed by Sophia Care Alliance™ in collaboration with artificial intelligence using HCOS™ Foundations, governance principles, disciplines, standards, and evidence-informed reasoning as guidance.

Human authorship, evidence verification, interpretation, revision, publication, and accountability remain essential.

Version Control

Future versions should record:

- Version number
- Publication date
- Revision date
- Document owner
- Human reviewers
- Technical reviewers
- Evidence updates
- Material changes
- Approval status
- Public feedback addressed
- Known limitations

Feedback and Revision

Feedback should be invited from:

- AI developers
- Researchers
- Practitioners
- Organizational leaders
- Privacy and security professionals
- Educators
- Policymakers
- Human factors specialists
- Affected workers
- Patients
- Students
- Communities affected by AI-supported systems

Feedback may identify:

- Errors
- Evidence gaps
- Safety concerns
- Privacy concerns
- Accessibility barriers
- Cultural limitations
- Governance weaknesses
- Unintended consequences
- Human load concerns
- Recommended revisions